Polar Coordinates & Polar Graphs

ANSWER KEY

Converting between polar and rectangular coordinates

1 Convert the polar point $\left(4, \frac{2\pi}{3}\right)$ to rectangular coordinates.
$x = r\cos\theta = 4\cos\left(\frac{2\pi}{3}\right) = 4\left(-\frac{1}{2}\right) = -2$. $y = r\sin\theta = 4\sin\left(\frac{2\pi}{3}\right) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$. Rectangular point: $(-2, 2\sqrt{3})$.

2 Convert the rectangular point $(-3, 3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.
$r = \sqrt{(-3)^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$. Since $(-3, 3)$ is in Quadrant II, $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$. Polar point: $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$.

Converting polar equations to rectangular form

3 Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.
Multiply both sides by r: $r^2 = 4r\cos\theta$, so $x^2 + y^2 = 4x$. Completing the square: $(x-2)^2 + y^2 = 4$ — a circle of radius 2 centered at $(2, 0)$.

4 Convert the rectangular equation $x^2 + y^2 = 9$ to polar form.
Since $r^2 = x^2 + y^2$, the equation becomes $r^2 = 9$, so $r = 3$ — a circle of radius 3 centered at the pole.

Visualizing a rectangular (r vs. θ) graph of a polar rose

5 The graph shows $r = 3\cos(2\theta)$ plotted in rectangular form, with θ on the horizontal axis (in radians) and r on the vertical axis — a standard technique for analyzing a polar rose curve before sketching it.

a State how many petals the actual polar rose $r = 3\cos(2\theta)$ has, using the rule for $r = a\cos(n\theta)$ with even n. When n is even, the rose $r = a\cos(n\theta)$ has $2n$ petals; here $n = 2$ gives 4 petals.

b Read off the graph: at which values of θ in $[0, 2\pi)$ does $r = 0$ (these are the angles where the rose curve passes through the pole)? $r = 0$ when $\cos(2\theta) = 0$, i.e. $2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$, so $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — matching the four zero-crossings shown.

Symmetry and special polar curves

6 Identify the type of curve represented by $r = 2 + 2\sin\theta$ and state its key feature.
This is a cardioid (heart-shaped curve), since it has the form $r = a + a\sin\theta$ (or $\cos\theta$) with equal constant and coefficient. It passes through the pole when $\sin\theta = -1$, i.e. $\theta = \frac{3\pi}{2}$.

7 Determine whether the graph of $r = 3\cos\theta$ is symmetric about the polar axis (the x-axis), and justify your answer.

Replacing θ with $-\theta$: $r = 3\cos(-\theta) = 3\cos\theta$, unchanged. So the graph is symmetric about the polar axis.

Multiple representations of the same polar point

8 Give two other polar representations of the point $\left(5, \frac{\pi}{4}\right)$: one with a negative r-value, and one with a different positive angle (adding 2π).

With negative r: $\left(-5, \frac{\pi}{4} + \pi\right) = \left(-5, \frac{5\pi}{4}\right)$. With added revolution: $\left(5, \frac{\pi}{4} + 2\pi\right) = \left(5, \frac{9\pi}{4}\right)$.

Applications of polar curves

9 A radar station models the range of its signal by the polar curve $r = 6 + 2\cos\theta$ (in km), where θ is measured from due east. Find the maximum and minimum detection range, and the direction of each.

Maximum range: $r = 6 + 2(1) = 8$ km, at $\theta = 0$ (due east). Minimum range: $r = 6 + 2(-1) = 4$ km, at $\theta = \pi$ (due west).

Distance between points in polar form

10 Find the distance between the polar points $\left(3, \frac{\pi}{6}\right)$ and $\left(5, \frac{2\pi}{3}\right)$ by first converting both to rectangular coordinates.

Point 1: $x = 3\cos\left(\frac{\pi}{6}\right) = 3\left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{2}$, $y = 3\sin\left(\frac{\pi}{6}\right) = \frac{3}{2}$. Point 2: $x = 5\cos\left(\frac{2\pi}{3}\right) = -\frac{5}{2}$, $y = 5\sin\left(\frac{2\pi}{3}\right) = \frac{5\sqrt{3}}{2}$. Distance $\approx \sqrt{(2.60 - (-2.5))^2 + (1.5 - 4.33)^2} \approx \sqrt{26.0 + 8.01} \approx 5.83$.

Identifying polar curve families

11 Classify each polar equation as a circle, cardioid, limaçon (with inner loop), or rose, and briefly justify: (a) $r = 5$, (b) $r = 2 + 4\sin\theta$, (c) $r = 4\sin(3\theta)$.

(a) $r = 5$ is a circle of radius 5 centered at the pole (constant r). (b) $r = 2 + 4\sin\theta$ has $|a| < |b|$ (here $a = 2 < b = 4$), so it is a limaçon with an inner loop. (c) $r = 4\sin(3\theta)$ has the rose form $r = a\sin(n\theta)$ with odd $n = 3$, giving a 3-petaled rose.

Points of intersection of polar curves

12 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 2$ and $r = 4\cos\theta$ intersect.

Set $2 = 4\cos\theta$: $\cos\theta = \frac{1}{2}$, giving $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$ (also written $-\frac{\pi}{3}$).

Area in polar coordinates (introductory)

13 Explain, without evaluating an integral, why the polar area formula $A = \frac{1}{2}\int_a^b r^2\,d\theta$ uses r^2 rather than r, by comparing it to the area of a circular sector $A = \frac{1}{2}r^2\theta$.

A circular sector of radius r and angle θ has area $\frac{1}{2}r^2\theta$ — note the r^2, since area scales with the square of the radius. The polar area formula generalizes this by summing infinitely many thin sectors, each with area $\frac{1}{2}r^2\,d\theta$, which is exactly why r (not r^2 or r^1 alone without justification) appears squared in the integrand.

14 A circular sector has radius 6 and central angle $\frac{\pi}{3}$. Use the sector-area formula $A = \frac{1}{2}r^2\theta$ to find its area.

$A = \frac{1}{2}(6)^2\left(\frac{\pi}{3}\right) = \frac{1}{2}(36)\left(\frac{\pi}{3}\right) = 6\pi \approx 18.85$.